Department of Computer Science and Engineering
UIET, CSJM University, Kanpur

C Programming Laboratory Assignment–4

Number Systems and Their Representation in C

Dr. Sandesh Gupta

What You Will Practise

Converting non-negative whole numbers between decimal, binary, octal and hexadecimal; checking conversions; using integer constants in C; and displaying one stored value in different number systems with `printf`.

Quick Revision

Binary (base 2)	Uses digits 0 and 1; place values are powers of 2.
Octal (base 8)	Uses digits 0 to 7; one octal digit represents three bits.
Hexadecimal (base 16)	Uses 0 to 9 and A to F; one hex digit represents four bits.
`25`, `031`, `0x19`	Decimal, octal and hexadecimal C constants for the same value.
`%d`, `%o`	Display an integer in decimal or octal.
`%x`, `%X`	Display hexadecimal using lowercase or uppercase letters.

Important Rules

Perform all work inside `~/C_Programming/Lab_04`. Convert each value yourself before using the computer. Do not use an online converter. No `scanf`, loop or decision statement is required in this assignment.

Preparation

Create and enter the directory `Lab_04`. Use `pwd` to confirm your location. Create a notebook table with the headings **Decimal**, **Binary**, **Octal** and **Hexadecimal**.

Problem 1: Decode the UIET Laboratory Number Cards

Situation

The UIET programming laboratory is testing number cards written in different systems. Your job is to find the value represented by every card.

Complete the missing entries. Show the place-value or division steps in your notebook; an answer without working is not sufficient.

Decimal	Binary	Octal	Hexadecimal
10			
	1111		
32			
	101101		
		77	
			A5
255			

For each completed row, convert your binary, octal or hexadecimal answer back to decimal. Circle any answer that failed this reverse check and correct it.

Be ready to explain: why 102_2 and 89_8 are invalid, and why 10_2, 10_8, 10_{10} and 10_{16} do not represent the same value.

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Problem 2: One Value, Four Representations

Situation

A C variable stores a value. The **printf** format decides how that value appears on the screen. You will use C as a checker for your manual conversions.

Create **representations.c** using this starting point:

```c
#include <stdio.h>

int main(void)
{
    int value = 45;

    printf("Decimal     = %d\n", value);
    printf("Octal       = %o\n", value);
    printf("Hexadecimal = %X\n", value);
    printf("Binary      = 101101\n");
    return 0;
}
```

1. Predict all four output lines before compiling.
2. Compile as **representations** and run it. Compare the output with your prediction.
3. Change **value** successively to 10, 15, 32 and 255. Before every run, manually replace the binary text with the correct binary form.
4. For each value, write the decimal, binary, octal and hexadecimal output in your notebook table.
5. Replace **%X** with **%x**. State exactly what changes and what does not change.

Important Observation

C's standard **printf** has formats for decimal, octal and hexadecimal, but no standard **%b** format for binary. In this beginner program, binary is written as text after you calculate it manually.

Problem 3: Different Constants, Same Stored Value

Situation

Three UIET computers receive the same identification number written in three different number systems. Prove through a C program that the stored values are equal.

Create **constants.c**. Declare these variables:

```c
int decimalNumber = 25;
int octalNumber   = 031;
int hexNumber     = 0x19;
```

Use three **printf** statements with **%d**. All three output lines should show decimal 25. Then display every variable once with **%o** and once with **%X**. Explain why changing the source notation does not create a different stored value.

After the correct program works, perform these experiments one at a time:

1. change **031** to **030** and predict its decimal value;
2. change **0x19** to **0x1A** and predict its decimal value;
3. try **int wrong = 08;** and read the first compiler error;
4. repair the program after every experiment.

Write in your notebook why **08** is invalid as an integer constant in C.

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Problem 4: Create Your UIET Number Identity Card

Choose a decimal value from 16 to 255. You may use the last two or three digits of your roll number if they fall in this range. Convert it manually to binary, octal and hexadecimal.

Create **number_card.c**. Store the chosen decimal value in one **int** variable. Produce an output similar to this:

```
----- UIET NUMBER IDENTITY CARD -----
Student       : Your Name
Decimal value : 45
Binary value  : 101101
Octal value   : 55
Hex value     : 2D
All four lines represent the same value.
```

Use the variable with **%d**, **%o** and **%X**; write the binary form as text. Compile, run and verify every line by converting it back to decimal.

Final Demonstration

Show Before Leaving the Lab

Show the instructor your completed conversion table, **representations.c**, **constants.c**, **number_card.c**, and their outputs. Demonstrate one change of value followed by saving, recompiling and running. Be prepared to explain one conversion and the difference among **%d**, **%o** and **%X**.

Remember: the value remains the same; only its representation changes.

SANDESH GUPTA
UIET
CSJM UNIVERSITY, KANPUR

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